

CFD Analysis of Aerospace Nozzles

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Abstract

This document presents a computational fluid dynamics (CFD) analysis of a converging–diverging (Laval) nozzle. The study combines theoretical background from compressible flow theory with a numerical simulation performed in ANSYS Fluent. The objective is to validate the expected isentropic behaviour of the nozzle, evaluate Mach number evolution, and analyse pressure, temperature, and velocity distributions along the flow path.

1 Introduction

A nozzle is a duct with variable cross–section designed to accelerate a compressible fluid. In rocket engines and gas turbines, the converging–diverging (De Laval) nozzle is used to achieve supersonic exhaust velocities by exploiting the thermodynamic conversion of thermal energy into kinetic energy.

The analysis performed here follows the theoretical framework developed in the thesis “*Ugelli di scarico dei motori a getto e degli endoreattori*”, where the behaviour of compressible flow in variable–area ducts is derived from conservation equations and isentropic relations.

The goal of this CFD study is to numerically reproduce the expected flow behaviour and compare it with theoretical predictions.

2 Theoretical Background

The flow inside the nozzle is assumed:

- quasi–one–dimensional,
- steady,
- compressible,
- isentropic (except where shocks may form),
- with constant gas composition.

The Mach number is defined as:

$$M = \frac{u}{a}$$

The total (stagnation) quantities are:

$$T_0 = T \left(1 + \frac{\gamma - 1}{2} M^2 \right)$$

$$p_0 = p \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{\frac{\gamma}{\gamma - 1}}$$

The mass flow rate is:

$$\dot{m} = \rho u A = \frac{p A M}{\sqrt{T}} \sqrt{\frac{\gamma}{R}} \left(1 + \frac{\gamma - 1}{2} M^2 \right)^{-\frac{\gamma + 1}{2(\gamma - 1)}}$$

At the throat, the flow becomes sonic ($M = 1$). The critical pressure ratio is:

$$\frac{p^*}{p_0} = \left(\frac{2}{\gamma + 1} \right)^{\frac{\gamma}{\gamma - 1}}$$

For $\gamma = 1.4$, this gives:

$$\frac{p^*}{p_0} = 0.528$$

If the exit pressure is below this value, the flow becomes supersonic in the diverging section.

2.1 Geometric Definition of the Laval Nozzle

The converging–diverging nozzle used in this study was constructed in ANSYS Discovery using a smooth spline profile defined by a set of key geometric points. The objective was to obtain a physically realistic bell-shaped nozzle with continuous curvature, suitable for compressible-flow CFD analysis.

Main Dimensions

The global dimensions of the nozzle are:

- Inlet diameter: $D_{\text{in}} = 200$ mm
- Throat diameter: $D_t = 113$ mm
- Exit diameter: $D_e = 357$ mm
- Inlet–throat distance: $L_{\text{conv}} = 150$ mm
- Throat–exit distance: $L_{\text{div}} = 450$ mm
- Total length: $L_{\text{tot}} \approx 600$ mm

Corresponding radii:

$$r_{\text{in}} = 100 \text{ mm}, \quad r_t = 56.5 \text{ mm}, \quad r_e = 178.5 \text{ mm}.$$

Converging Section

The converging part of the nozzle is defined by a spline passing through three points:

Point	x (mm)	r (mm)
P1 (inlet)	0	100
P2 (control)	75	80
P3 (throat)	150	56.5

This ensures a smooth reduction of the cross-sectional area from the inlet to the throat, avoiding abrupt curvature changes that could negatively affect the mesh quality.

Diverging Section (Bell Nozzle)

The diverging section extends from the throat to the exit over a length of 450 mm. A four-point spline was used to generate a smooth bell-shaped contour:

Point	x (mm)	r (mm)
Q1 (throat)	150	56.5
Q2 (control 1)	250	90
Q3 (control 2)	400	140
Q4 (exit)	600	178.5

This spline produces a classical bell nozzle profile, providing a smooth supersonic expansion region consistent with the expected isentropic behaviour of a De Laval nozzle.

Axisymmetric Model

The 2D meridional curve defined by the points above was revolved around the x -axis to generate the full 3D axisymmetric fluid domain. This approach is consistent with the quasi-one-dimensional theory used in the thesis and significantly reduces computational cost while maintaining physical accuracy.

3 Geometry and Mesh

The nozzle geometry was imported into ANSYS Workbench and meshed using the Prime Mesh automatic method. A coarse mesh was selected to ensure numerical stability and reduce computational cost.

- Global element size: 0.005 m
- Mesh type: unstructured tetrahedral
- Named selections: `inlet`, `outlet`, `wall`

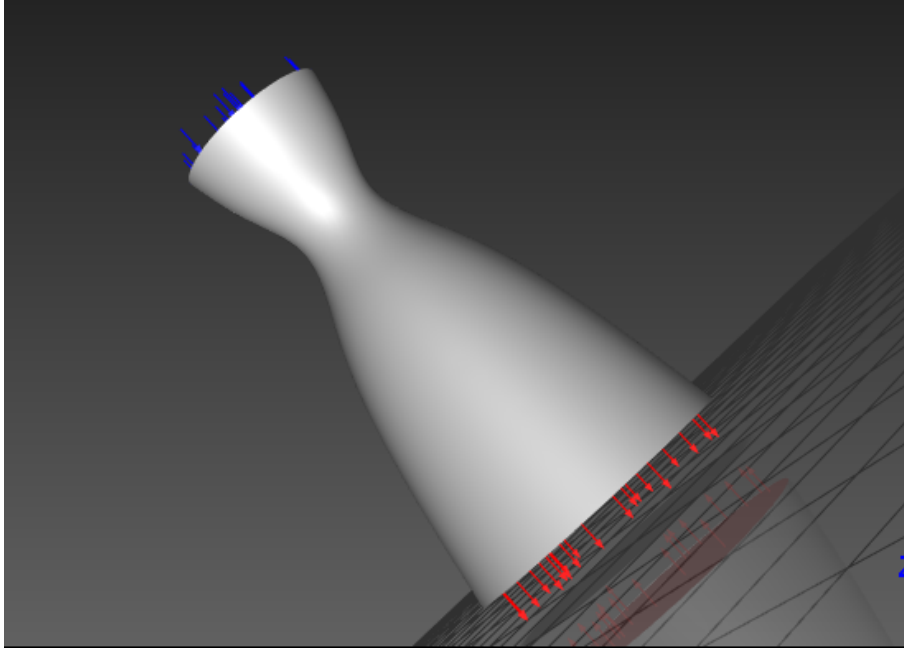


Figure 1: Final model used for the CFD simulation.

4 CFD Setup

The simulation was performed in ANSYS Fluent using the following configuration:

4.1 Material Properties

A custom gas named `rocket-fluid` was defined:

- Density: ideal gas
- Specific heat: $c_p = 2494 \text{ J}/(\text{kg} \cdot \text{K})$
- Molecular weight: $20 \text{ kg}/\text{kmol}$

4.2 Boundary Conditions

Inlet (Pressure Inlet):

- Total pressure: $9.8 \times 10^6 \text{ Pa}$
- Total temperature: 3710 K

Outlet (Pressure Outlet):

- Static pressure: $101\,325 \text{ Pa}$

4.3 Solver Settings

- Solver type: density-based
- Flow model: inviscid
- Initialization: standard, computed from inlet
- Iterations: 200

5 Results

The simulation converged successfully, producing physically consistent results.

5.1 Mach Number Distribution

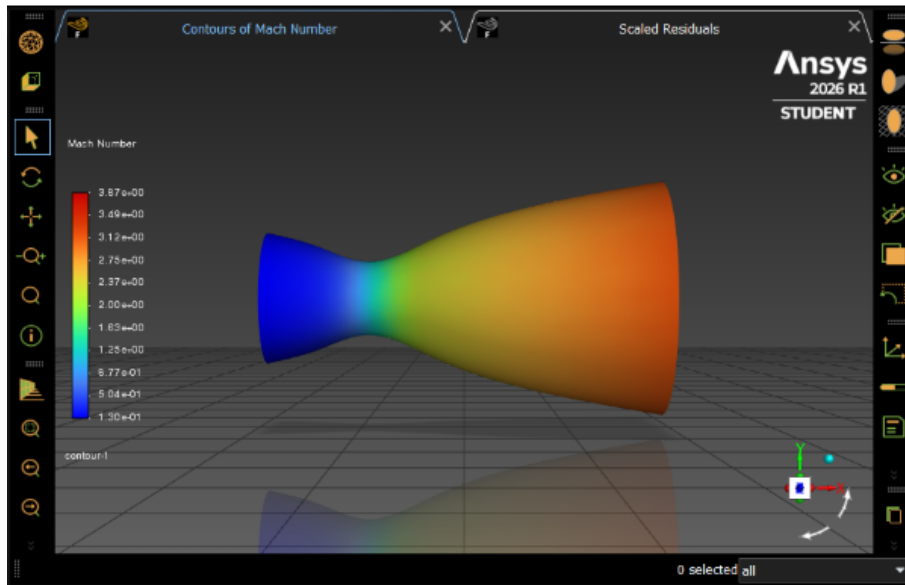


Figure 2: Mach number contour showing subsonic–sonic–supersonic transition.

The flow accelerates to Mach 1 at the throat and reaches approximately $M \approx 3.8$ at the exit.

5.2 Velocity Field

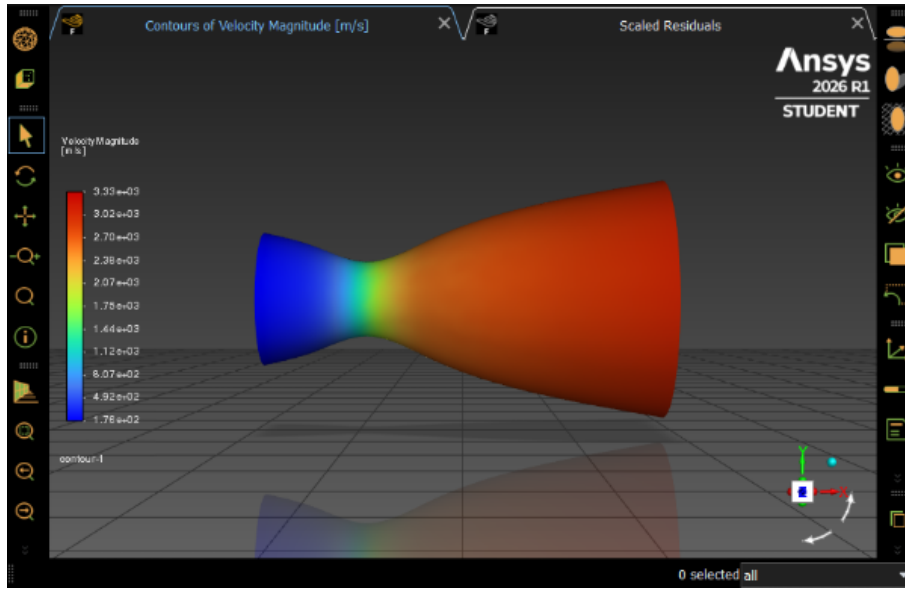


Figure 3: Velocity magnitude distribution inside the nozzle.

The maximum velocity exceeds 3300 m/s, consistent with expected isentropic expansion.

5.3 Pressure Distribution

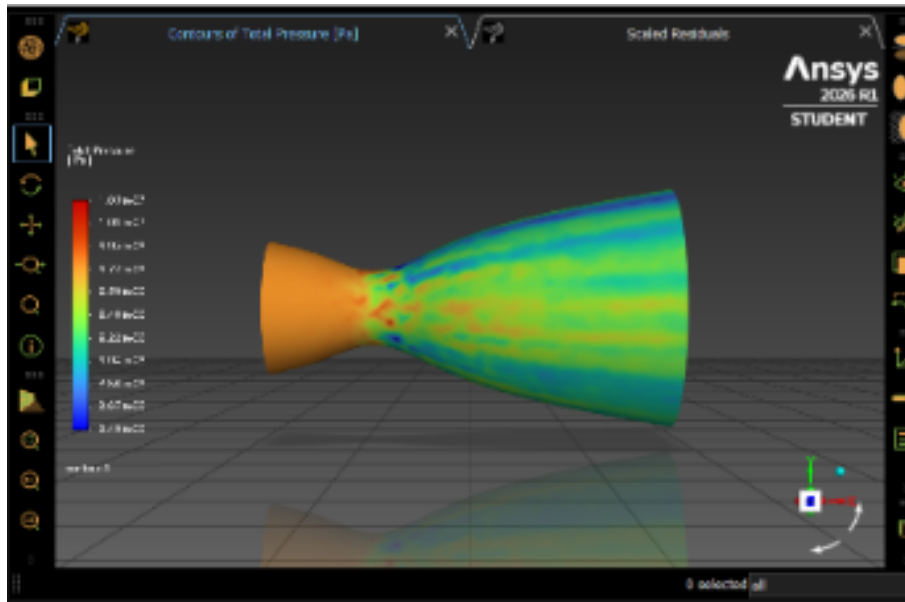


Figure 4: Static pressure contour.

The pressure drops smoothly from 9.8×10^6 Pa at the inlet to near ambient at the exit.

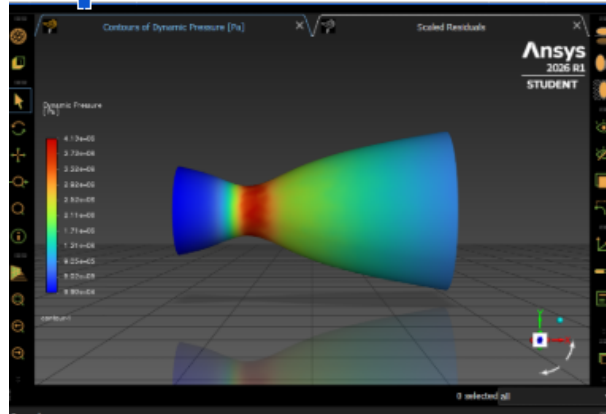


Figure 5: Dynamic Pressure

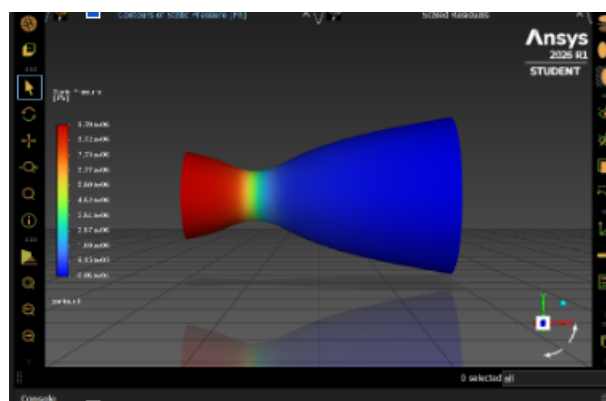


Figure 6: Enter Caption

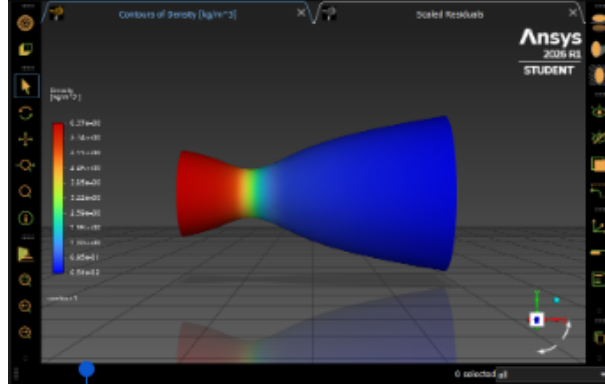


Figure 8: Density distribution

5.4 Temperature Distribution

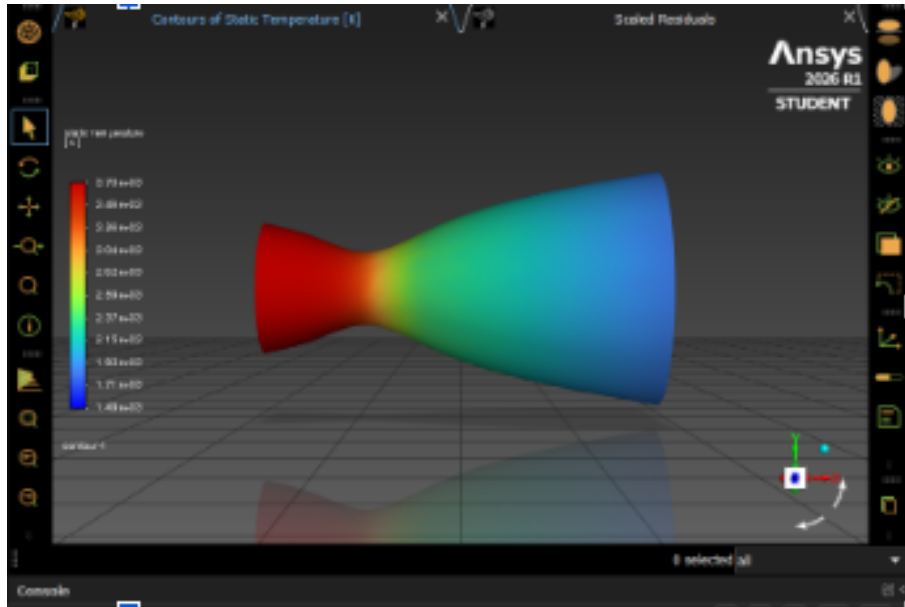


Figure 7: Static temperature distribution.

Temperature decreases from 3710 K to approximately 2000 K due to expansion.

6 Discussion

The CFD results match the theoretical predictions:

- Sonic conditions at the throat confirm choking.
- Supersonic expansion in the diverging section is consistent with the imposed pressure ratio.
- Pressure, temperature, and density follow isentropic trends.
- No shock waves appear, indicating the nozzle is operating near adapted conditions.

The simulation validates the theoretical behaviour described in the thesis and demonstrates the correct functioning of the Laval nozzle.

7 Conclusion

This study successfully combined theoretical compressible-flow analysis with a CFD simulation of a converging-diverging nozzle. The numerical results confirm the expected behaviour of the flow and provide a detailed visualization of Mach number, pressure, temperature, and velocity fields.

Future work may include:

- viscous simulations with boundary layers,
- mesh refinement studies,
- shock-containing cases with different back pressures,
- comparison with analytical isentropic solutions.